HARESH PANCHAL

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Professional Summary

Embedded Software/Firmware Engineer with **9.5+ years of experience** in embedded firmware development, DSP, microcontrollers, communication protocols, and desktop applications. Experienced in **Embedded C/C++, RTOS, Linux, automotive and industrial applications**. Proficient in **Agile development, JIRA, and Git version control**.

Technical Skills

- **Programming Languages:** Embedded C, C++ 11/14/17/20, STL, Multi-threading, Data Structure, Design Pattern, Python C#
- **Linux Programming:** Linux Device Driver, Kernal Programming, Basic File I/O, Threads, Signal handling, Memory Management, IPC Mechanism
- **Operating System:** Windows, Linux, RTOS.
- **Tools:** CCSv8, MPLAB, e2Studio, CS+, Keil, IAR, PyCharm, VS Code, Visual Studio, Eclipse, VI editor, GCC Compiler, GDB Tool, Turbo C, Qt Creator.
- **Protocol & Peripheral:** I2C, SPI, UART, CAN, CAN-FD, MODBUS RTU/TCP, Ethernet, PWM, FSI, ECAP, ADC, DAC, RTC, EEPROM, LED, LCD.

Work Experience

❖ **Accenture Business Solution Limited** (May 2024 to Present)

1. **Development of testbench system to test vehicle functionality**
   - Developed modular peripheral software for **CAN, CAN FD, and I/O expander interfaces**.
   - Designed and developed CAN communication software using the MCP2517 CAN FD controller with **epoll-driven asynchronous event processing** and comprehensive **CAN bus error management**.
   - Implemented interface for slave devices using observer design pattern
   - Implemented a **modular hardware abstraction layer (HAL)** for MAX7300-based GPIO expansion.
   - Developed user-space software the ADSD3500 camera sensor. Integrated and tested CAN and camera modules with the system.
   - Implemented systemd services for peripheral management and system health monitoring
   - Developed RESTful APIs to enable efficient frontend– backend communication.

❖ **Wipro Limited** (Dec 2021 to May 2024)

1. Ford Motor (HMI Middleware Development)

- Worked on middleware of Drive Assist feature like Active Park Assist, Park Distance Control, Trailer back up Assist, Auto Hitch, and Reverse Video Camera for HMI Development using C++11.
- Worked on root cause analysis based on FNV and CAN log.

➢ **Rewards: Developed CAN log analyser tool.**

- Similar like Canoe.
- Extract signals values from log file (.asc or blf) and connect with data base file (DBC).
- Plot multiple signals simultaneous;

❖ **AMTECH Electronics India Pvt Ltd** (Aug 2018 to Dec 2021)

1. Automated Test Bench for VFD Control & Driver Cards

- Developed an automated test and calibration system using **TI TMS320F28377D, Embedded C, C++, C#**, and **Modbus RTU/TCP**.
- Implemented PWM, power supply, DC voltage, analog I/O, ADC, digital I/O, gate pulse, and short-circuit tests.
- Developed PC GUI for test execution, monitoring, and automated report generation.
- **Achievement:** Eliminated manual testing of control and driver cards.

2. VFD Seven-Segment & Graphical Display

- Designed and developed VFD display and keypad interface using **Renesas MCU and Embedded C**.
- Implemented **Modbus Master** communication and parameter management for **2500+ VFD parameters**.

3. Analog & Digital Fiber Optic Communication

- Developed analog/digital communication over **FSI** using **TI TMS320F28377D and Embedded C**.
- Utilized **CLA** for high-speed signal processing.
- **Achievement:** Achieved **500 kHz bandwidth with <0.01% linearity error**.

4. Customized Modbus Communication DLL

- Developed **C++ DLL** for customized **Modbus RTU and Modbus TCP** communication.
- Integrated communication interface with PC-based applications.

5. VFD GUI / Desktop Application Development

- Developed **C#/C++ applications using Visual Studio 2019** for VFD monitoring and configuration.
- Implemented Modbus RTU/TCP Master, parameter monitoring, graphs, fault display, and configuration import/export.

6. TI TMS320F28388D Dual-Core DSP

- Explored and developed communication/peripheral applications using **TMS320F28388D**.
- Worked with **Modbus, Modbus TCP, SPI, I2C, CAN, FSI, and Analog/Digital I/O**.

❖ **Ciright Hardware System Pvt Ltd.** (Jan 2017 to May 2018)

1. **Smart 4-Channel Wi-Fi Switch**
   - **Software:** IAR, Embedded C
   - **Hardware:** STM32F051, ESP8266
   - Developed a 4-channel, 16A smart switch controlled through an Android application.
   - Implemented ON/OFF pulse control and scheduling functionality for each channel.

Training

❖ **Vector India Limited, Bangalore.** (Mar 2016 to Sep 2016)
- Completed professional training in Embedded Systems, C, C++, Linux, Shell Scripting, and RTOS.

Embedded Training Project

Data Acquisition System using 8051
- Developed a data acquisition system using **8051** with **I2C, SPI, and UART** communication.
- Interfaced **DS1307 RTC** and **DS1621 digital temperature sensor**.
- Displayed sensor data on **LCD and HyperTerminal**.
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Education Qualification

❖ B.E. Electrical from GTU in 2013